Supplementary Material B for Theorem 36

On the Minimum Distance of High-rate Moore-Type Quasi-Cyclic LDPC Codes

Detailed Derivations for Weight-8 Cases II–VII

1 Scope

Throughout, let m be an odd prime and $a, b \in \mathbb{Z}_m^*$ satisfy
$\mathrm{ord}_m(a) = k > 3$ and $\mathrm{ord}_m(b) = 3$.
This supplement derives Cases II–VII of Theorem 36 for Moore$(m, a, b, 3)$.
Case I is proved in the main manuscript; its notation and conditions (a)–(d) are retained.

The correspondence between the case labels used below and the matching subcases is

Case	Matching subcase	RREF	Free variables
II	1-2-1	A_{121}	u_4, u_7, u_8
III	1-2-2	A_{122}	u_6, u_7, u_8
IV	2-1-2	A_{212}	u_4, u_6, u_7, u_8
V	2-1-7	A_{217}	u_4, u_6, u_7, u_8
VI	2-3-5	A_{235}	u_4, u_6, u_7, u_8
VII	2-4-3	A_{243}	u_5, u_6, u_7, u_8

2 Algebraic framework and conventions

We identify $\mathbb{Z}_m$ with $\mathbb{F}_m$; all calculations are carried out in $\mathbb{F}_m$.
We use the normalization $c_1 = c(0, 0)$ and write

$$u_i = a^{t_i}, \quad t_i \in \mathbb{Z}_k, \quad u_1 = 1.$$

Since $\mathrm{ord}_m(b) = 3$, we have $b \neq 1$ and

$$1 + b + b^2 = 0.$$

All coefficients below are simplified using this identity.

For a matching edge $\{c_i, c_q\}$ at level $s \in \{0, 1, 2\}$, the basic relation is

$$x_i - x_q = (b^s - 1)(u_q - u_i).$$

For an oriented cycle

$$\underline{i_1}, \underline{i_2}, \ldots, \underline{i_\ell}, \underline{i_1}(s_1, \ldots, s_\ell),$$

the underlined indices denote the successive vertices and s_r is the level of the edge joining c_{i_r} to $c_{i_{r+1}}$, with $i_{\ell+1} = i_1$. Summing the basic edge relations around the cycle eliminates the x-coordinates and gives a homogeneous linear equation in $u_1, \ldots, u_8$.

For each configuration, choose a spanning tree and form the fundamental-cycle equations. These equations are sufficient as well as necessary: the tree-edge relations determine all x_i from

one root value, and each non-tree edge is compatible with them exactly when its fundamental-cycle equation vanishes. We then row-reduce in the fixed column order $(u_1, \dots, u_8)$, omitting zero rows. After $u_1 = 1$, the pivot relation for u_1 gives condition *(b)*, and the remaining dependent variables give condition *(d)* through membership in $\langle a \rangle$.

For distinctness, the level-0 blocks must have different common x-coordinates, while columns inside the same level-0 block must have different exponents. Substituting the RREF relations and the main congruence into all required differences gives conditions *(a)* and *(c)*. Factors differing by a nonzero scalar in $\mathbb{F}_m$ are identified.

Labeling convention for Case VI. For subcase 2-3-5, we use the representative in Proposition 29. Relative to the intermediate enumeration label, it is obtained by $c_7 \leftrightarrow c_8$; this fixes the column order used for A_{235}.

3 Case II: Subcase 1-2-1

3.1 Matching configuration and fundamental-cycle system

The matching set is

$$\begin{aligned} M_0 &= \{\{c_1, c_2\},\ \{c_3, c_4\},\ \{c_5, c_6\},\ \{c_7, c_8\}\}, \\ M_1 &= \{\{c_1, c_3, c_6, c_8\},\ \{c_2, c_5\},\ \{c_4, c_7\}\}, \\ M_2 &= \{\{c_1, c_4\},\ \{c_2, c_3\},\ \{c_5, c_8\},\ \{c_6, c_7\}\}. \end{aligned}$$

We choose the spanning tree

$$F = \{c_1, c_3\}_1,\ \{c_1, c_6\}_1,\ \{c_1, c_8\}_1,\ \{c_2, c_5\}_1,\ \{c_4, c_7\}_1,\ \{c_1, c_4\}_2,\ \{c_2, c_3\}_2.$$

This tree has 7 edges. Since the matching graph has 16 edges, it determines 9 fundamental cycles. The associated cycle equations are listed below. Three fundamental cycles, $\underline{3,1,6}(111)$, $\underline{3,1,8}(111)$, and $\underline{6,1,8}(111)$, lie entirely in the level-1 4-matching and give the identity $0 = 0$. The six nontrivial fundamental-cycle equations are:

(1) $\underline{5,2,3,1,8}(12112)$:

$$(1+2b)\,u_2 + (-1-2b)\,u_3 + (-1-2b)\,u_5 + (1+2b)\,u_8 = 0.$$

(2) $\underline{6,1,4,7}(1212)$:

$$(1+2b)\,u_1 + (-1-2b)\,u_4 + (-1-2b)\,u_6 + (1+2b)\,u_7 = 0.$$

(3) $\underline{1,3,2}(120)$:

$$(1-b)\,u_1 + (-2-b)\,u_2 + (1+2b)\,u_3 = 0.$$

(4) $\underline{3,1,4}(120)$:

$$(1+2b)\,u_1 + (1-b)\,u_3 + (-2-b)\,u_4 = 0.$$

(5) $\underline{5,2,3,1,6}(12110)$:

$$(1+2b)\,u_2 + (-1-2b)\,u_3 + (1-b)\,u_5 + (-1+b)\,u_6 = 0.$$

(6) $\underline{7,4,1,8}(1210)$:

$$(-1-2b)\,u_1 + (1+2b)\,u_4 + (1-b)\,u_7 + (-1+b)\,u_8 = 0.$$

After deleting zero equations and row-reducing the remaining system in the column order $(u_1, \ldots, u_8)$, we obtain

$$A_{121} = \begin{bmatrix} 1 & 0 & 0 & -1 & 0 & 0 & 1+b & -1-b \\ 0 & 1 & 0 & -1 & 0 & 0 & 2+b & -2-b \\ 0 & 0 & 1 & -1 & 0 & 0 & 1 & -1 \\ 0 & 0 & 0 & 0 & 1 & 0 & 1+b & -2-b \\ 0 & 0 & 0 & 0 & 0 & 1 & b & -1-b \end{bmatrix}.$$

Hence this is the RREF of the fundamental-cycle system.

3.2 RREF solution and conditions *(b)* and *(d)*

The free variables are

$$u_4,\ u_7,\ u_8.$$

The pivot variables are therefore

$$\begin{aligned} u_1 &= u_4 + (-1-b)\,u_7 + (1+b)\,u_8, \\ u_2 &= u_4 + (-2-b)\,u_7 + (2+b)\,u_8, \\ u_3 &= u_4 - u_7 + u_8, \\ u_5 &= (-1-b)\,u_7 + (2+b)\,u_8, \\ u_6 &= -bu_7 + (1+b)\,u_8. \end{aligned}$$

Imposing $u_1 = 1$ gives the main congruence

$$-1 + u_4 + (-1-b)\,u_7 + (1+b)\,u_8 = 0,$$

which is condition *(b)* after substituting $u_i = a^{t_i}$. Requiring the remaining dependent variables above to belong to $\langle a \rangle$ gives condition *(d)*.

3.3 Distinctness conditions *(a)* and *(c)*

For the distinctness calculation we solve the main congruence for u_4:

$$u_4 = 1 + (1+b)\,u_7 + (-1-b)\,u_8.$$

After this substitution, the common x-coordinate of each block of M_0 is

$$\begin{aligned} x_1 &= x_2 = 0, \\ x_3 &= x_4 = (1+2b)\,u_7 + (-1-2b)\,u_8, \\ x_5 &= x_6 = -1 + b + (-1-2b)\,u_7 + (2+b)\,u_8, \\ x_7 &= x_8 = -1 + b + (1-b)\,u_8. \end{aligned}$$

Hence different level-0 blocks have distinct x-coordinates, and the columns inside each block are distinct, exactly when the following factors are all nonzero (factors differing by a nonzero scalar are identified):

$$\begin{gathered} u_7 - u_8 \not\equiv 0 \pmod{m}, \\ 1 + bu_7 + (-1-b)\,u_8 \not\equiv 0 \pmod{m}, \\ 1 - u_8 \not\equiv 0 \pmod{m}, \\ 1 + 2bu_7 + (-1-2b)\,u_8 \not\equiv 0 \pmod{m}. \end{gathered}$$

Using the main congruence, these factors reduce to condition *(a)* together with the nonzero factors in condition *(c)*. Explicitly,

$$\begin{aligned}
\text{a1:}\quad & t_7 \neq t_8 \Longleftrightarrow u_7 - u_8 \not\equiv 0 \pmod m,\\
\text{a2:}\quad & t_8 \neq 0 \Longleftrightarrow 1 - u_8 \not\equiv 0 \pmod m,\\
& \text{c1} \Longleftrightarrow 1 + bu_7 + (-1-b)\,u_8 \not\equiv 0 \pmod m,\\
& \text{c2} \Longleftrightarrow 1 + 2bu_7 + (-1-2b)\,u_8 \not\equiv 0 \pmod m.
\end{aligned}$$

Hence conditions *(a)* and *(c)* are precisely the level-0 and within-block distinctness requirements for this matching configuration. Together with conditions *(b)* and *(d)*, they give the necessary and sufficient algebraic conditions for this case.

4 Case III: Subcase 1-2-2

4.1 Matching configuration and fundamental-cycle system

The matching set is

$$\begin{aligned}
M_0 &= \{\{c_1, c_2\},\ \{c_3, c_4, c_5, c_6\},\ \{c_7, c_8\}\},\\
M_1 &= \{\{c_1, c_3\},\ \{c_2, c_5\},\ \{c_4, c_7\},\ \{c_6, c_8\}\},\\
M_2 &= \{\{c_1, c_4\},\ \{c_2, c_6\},\ \{c_3, c_7\},\ \{c_5, c_8\}\}.
\end{aligned}$$

We choose the spanning tree

$$F = \{c_1, c_3\}_1,\ \{c_2, c_5\}_1,\ \{c_4, c_7\}_1,\ \{c_6, c_8\}_1,\ \{c_1, c_4\}_2,\ \{c_2, c_6\}_2,\ \{c_1, c_2\}_0.$$

This tree has 7 edges. Since the matching graph has 16 edges, it determines 9 fundamental cycles. The associated cycle equations are listed below.

(1) $\underline{3,1,4,7}(1212)$:

$$(1+2b)\,u_1 + (-1-2b)\,u_3 + (-1-2b)\,u_4 + (1+2b)\,u_7 = 0.$$

(2) $\underline{5,2,6,8}(1212)$:

$$(1+2b)\,u_2 + (-1-2b)\,u_5 + (-1-2b)\,u_6 + (1+2b)\,u_8 = 0.$$

(3) $\underline{3,1,4}(120)$:

$$(1+2b)\,u_1 + (1-b)\,u_3 + (-2-b)\,u_4 = 0.$$

(4) $\underline{3,1,2,5}(1010)$:

$$(-1+b)\,u_1 + (1-b)\,u_2 + (1-b)\,u_3 + (-1+b)\,u_5 = 0.$$

(5) $\underline{3,1,2,6}(1020)$:

$$(-1+b)\,u_1 + (2+b)\,u_2 + (1-b)\,u_3 + (-2-b)\,u_6 = 0.$$

(6) $\underline{4,1,2,5}(2010)$:

$$(-2-b)\,u_1 + (1-b)\,u_2 + (2+b)\,u_4 + (-1+b)\,u_5 = 0.$$

(7) $\underline{4,1,2,6}(2020)$:

$$(-2-b)\,u_1 + (2+b)\,u_2 + (2+b)\,u_4 + (-2-b)\,u_6 = 0.$$

(8) $\underline{5,2,6}(120)$:

$$(1+2b)\,u_2 + (1-b)\,u_5 + (-2-b)\,u_6 = 0.$$

(9) $\underline{7,4,1,2,6,8}(120210)$:

$$\begin{aligned}(-2-b)\,u_1 + (2+b)\,u_2 + (1+2b)\,u_4 + (-1-2b)\,u_6 \\ + (1-b)\,u_7 + (-1+b)\,u_8 = 0.\end{aligned}$$

After deleting zero equations and row-reducing the remaining system in the column order $(u_1, \ldots, u_8)$, we obtain

$$A_{122} = \begin{bmatrix} 1 & 0 & 0 & 0 & 0 & -1+b & -1 & 1-b \\ 0 & 1 & 0 & 0 & 0 & -1+b & 0 & -b \\ 0 & 0 & 1 & 0 & 0 & b & -1 & -b \\ 0 & 0 & 0 & 1 & 0 & -1 & -1 & 1 \\ 0 & 0 & 0 & 0 & 1 & b & 0 & -1-b \end{bmatrix}.$$

Hence this is the RREF of the fundamental-cycle system.

4.2 RREF solution and conditions *(b)* and *(d)*

The free variables are

$$u_6,\ u_7,\ u_8.$$

The pivot variables are therefore

$$\begin{aligned} u_1 &= (1-b)\,u_6 + u_7 + (-1+b)\,u_8, \\ u_2 &= (1-b)\,u_6 + bu_8, \\ u_3 &= -bu_6 + u_7 + bu_8, \\ u_4 &= u_6 + u_7 - u_8, \\ u_5 &= -bu_6 + (1+b)\,u_8. \end{aligned}$$

Imposing $u_1 = 1$ gives the main congruence

$$-1 + (1-b)\,u_6 + u_7 + (-1+b)\,u_8 = 0,$$

which is condition *(b)* after substituting $u_i = a^{t_i}$. Requiring the remaining dependent variables above to belong to $\langle a \rangle$ gives condition *(d)*.

4.3 Distinctness conditions *(a)* and *(c)*

For the distinctness calculation we solve the main congruence for u_7:

$$u_7 = 1 + (-1+b)\,u_6 + (1-b)\,u_8.$$

After this substitution, the common x-coordinate of each block of M_0 is

$$\begin{aligned} x_1 = x_2 &= 0, \\ x_3 = x_4 = x_5 = x_6 &= (-1+b)\,u_6 + (1-b)\,u_8, \\ x_7 = x_8 &= (-2+2b)\,u_6 + (2-2b)\,u_8. \end{aligned}$$

Hence different level-0 blocks have distinct x-coordinates, and the columns inside each block are distinct, exactly when the following factors are all nonzero (factors differing by a nonzero scalar are identified):

$$\begin{gathered} u_6 - u_8 \not\equiv 0 \pmod m, \\ 1 + (-1+b)\,u_6 - bu_8 \not\equiv 0 \pmod m, \\ 1 - 2u_6 + u_8 \not\equiv 0 \pmod m, \\ 1 + 2bu_6 + (-1-2b)\,u_8 \not\equiv 0 \pmod m. \end{gathered}$$

Using the main congruence, these factors reduce to condition *(a)* together with the nonzero factors in condition *(c)*. Explicitly,

$$\begin{aligned} \text{a1:}\quad & t_6 \neq t_8 \Longleftrightarrow u_6 - u_8 \not\equiv 0 \pmod m, \\ \text{a2:}\quad & t_7 \neq t_8 \Longleftrightarrow 1 + (-1+b)\,u_6 - bu_8 \not\equiv 0 \pmod m, \\ & \text{c1} \Longleftrightarrow 1 - 2u_6 + u_8 \not\equiv 0 \pmod m, \\ & \text{c2} \Longleftrightarrow 1 + 2bu_6 + (-1-2b)\,u_8 \not\equiv 0 \pmod m. \end{aligned}$$

Hence conditions *(a)* and *(c)* are precisely the level-0 and within-block distinctness requirements for this matching configuration. Together with conditions *(b)* and *(d)*, they give the necessary and sufficient algebraic conditions for this case.

5 Case IV: Subcase 2-1-2

5.1 Matching configuration and fundamental-cycle system

The matching set is

$$\begin{aligned} M_0 &= \{\{c_1,c_2\},\ \{c_3,c_4\},\ \{c_5,c_6\},\ \{c_7,c_8\}\}, \\ M_1 &= \{\{c_1,c_3\},\ \{c_2,c_4\},\ \{c_5,c_7\},\ \{c_6,c_8\}\}, \\ M_2 &= \{\{c_1,c_5\},\ \{c_3,c_6\},\ \{c_2,c_7\},\ \{c_4,c_8\}\}. \end{aligned}$$

We choose the spanning tree

$$F = \{c_1,c_3\}_1,\ \{c_2,c_4\}_1,\ \{c_5,c_7\}_1,\ \{c_6,c_8\}_1,\ \{c_1,c_5\}_2,\ \{c_2,c_7\}_2,\ \{c_3,c_6\}_2.$$

This tree has 7 edges. Since the matching graph has 12 edges, it determines 5 fundamental cycles. The associated cycle equations are listed below.

(1) $\underline{4,2,7,5,1,3,6,8}$(12121212):

$$\begin{aligned} &(-1-2b)\,u_1 + (1+2b)\,u_2 + (1+2b)\,u_3 + (-1-2b)\,u_4 \\ &\quad + (1+2b)\,u_5 + (-1-2b)\,u_6 + (-1-2b)\,u_7 + (1+2b)\,u_8 = 0. \end{aligned}$$

(2) $\underline{1,5,7,2}$(2120):

$$(2+b)\,u_1 + (-2-b)\,u_2 + (-1-2b)\,u_5 + (1+2b)\,u_7 = 0.$$

(3) $\underline{3,1,5,7,2,4}$(121210):

$$\begin{aligned} &(1+2b)\,u_1 + (-1-2b)\,u_2 + (1-b)\,u_3 + (-1+b)\,u_4 \\ &\quad + (-1-2b)\,u_5 + (1+2b)\,u_7 = 0. \end{aligned}$$

(4) $\underline{5,1,3,6}$(2120):

$$(-1-2b)\,u_1 + (1+2b)\,u_3 + (2+b)\,u_5 + (-2-b)\,u_6 = 0.$$

(5) $\underline{7,5,1,3,6,8}(121210)$:

$$\begin{aligned}(-1-2b)\,u_1+(1+2b)\,u_3+(1+2b)\,u_5+(-1-2b)\,u_6\\ +(1-b)\,u_7+(-1+b)\,u_8=0.\end{aligned}$$

After deleting zero equations and row-reducing the remaining system in the column order $(u_1,\dots,u_8)$, we obtain

$$A_{212}=\begin{bmatrix}1&0&0&-1&0&-1-b&b&1\\0&1&0&-1&0&0&b&-b\\0&0&1&-1&0&-1-b&0&1+b\\0&0&0&0&1&-1&-1&1\end{bmatrix}.$$

Hence this is the RREF of the fundamental-cycle system.

5.2 RREF solution and conditions *(b)* and *(d)*

The free variables are

$$u_4,\ u_6,\ u_7,\ u_8.$$

The pivot variables are therefore

$$\begin{aligned}u_1&=u_4+(1+b)\,u_6-bu_7-u_8,\\ u_2&=u_4-bu_7+bu_8,\\ u_3&=u_4+(1+b)\,u_6+(-1-b)\,u_8,\\ u_5&=u_6+u_7-u_8.\end{aligned}$$

Imposing $u_1=1$ gives the main congruence

$$-1+u_4+(1+b)\,u_6-bu_7-u_8=0,$$

which is condition *(b)* after substituting $u_i=a^{t_i}$. Requiring the remaining dependent variables above to belong to $\langle a\rangle$ gives condition *(d)*.

5.3 Distinctness conditions *(a)* and *(c)*

For the distinctness calculation we solve the main congruence for u_4:

$$u_4=1+(-1-b)\,u_6+bu_7+u_8.$$

After this substitution, the common x-coordinate of each block of M_0 is

$$\begin{aligned}x_1&=x_2=0,\\ x_3&=x_4=(1+2b)\,u_7+(-1-2b)\,u_8,\\ x_5&=x_6=-2-b+(2+b)\,u_6+(2+b)\,u_7+(-2-b)\,u_8,\\ x_7&=x_8=-2-b+(1+2b)\,u_6+(2+b)\,u_7+(-1-2b)\,u_8.\end{aligned}$$

Hence different level-0 blocks have distinct x-coordinates, and the columns inside each block are distinct, exactly when the following factors are all nonzero (factors differing by a nonzero scalar are identified):

$$\begin{gathered}u_7-u_8\not\equiv 0\pmod m,\\ 1-u_6-u_7+u_8\not\equiv 0\pmod m,\\ 1+(-1-b)\,u_6-u_7+(1+b)\,u_8\not\equiv 0\pmod m,\\ 1-u_6+bu_7-bu_8\not\equiv 0\pmod m,\\ 1+(-1-b)\,u_6+bu_7\not\equiv 0\pmod m,\\ u_6-u_8\not\equiv 0\pmod m.\end{gathered}$$

Using the main congruence, these factors reduce to condition *(a)* together with the nonzero factors in condition *(c)*. Explicitly,

$$\begin{aligned}
\text{a1:}\quad & t_6 \neq t_8 \Longleftrightarrow u_6 - u_8 \not\equiv 0 \pmod m,\\
\text{a2:}\quad & t_7 \neq t_8 \Longleftrightarrow u_7 - u_8 \not\equiv 0 \pmod m,\\
& \text{c1} \Longleftrightarrow 1 - u_6 - u_7 + u_8 \not\equiv 0 \pmod m,\\
& \text{c2} \Longleftrightarrow 1 + (-1-b)\,u_6 - u_7 + (1+b)\,u_8 \not\equiv 0 \pmod m,\\
& \text{c3} \Longleftrightarrow 1 - u_6 + bu_7 - bu_8 \not\equiv 0 \pmod m,\\
& \text{c4} \Longleftrightarrow 1 + (-1-b)\,u_6 + bu_7 \not\equiv 0 \pmod m.
\end{aligned}$$

Hence conditions *(a)* and *(c)* are precisely the level-0 and within-block distinctness requirements for this matching configuration. Together with conditions *(b)* and *(d)*, they give the necessary and sufficient algebraic conditions for this case.

6 Case V: Subcase 2-1-7

6.1 Matching configuration and fundamental-cycle system

The matching set is

$$\begin{aligned}
M_0 &= \{\{c_1,c_2\},\ \{c_3,c_4\},\ \{c_5,c_6\},\ \{c_7,c_8\}\},\\
M_1 &= \{\{c_1,c_3\},\ \{c_2,c_4\},\ \{c_5,c_7\},\ \{c_6,c_8\}\},\\
M_2 &= \{\{c_1,c_5\},\ \{c_2,c_6\},\ \{c_3,c_7\},\ \{c_4,c_8\}\}.
\end{aligned}$$

We choose the spanning tree

$$F = \{c_1,c_3\}_1,\ \{c_2,c_4\}_1,\ \{c_5,c_7\}_1,\ \{c_6,c_8\}_1,\ \{c_1,c_5\}_2,\ \{c_2,c_6\}_2,\ \{c_1,c_2\}_0.$$

This tree has 7 edges. Since the matching graph has 12 edges, it determines 5 fundamental cycles. The associated cycle equations are listed below.

(1) $\underline{3,1,5,7}(1212)$:

$$(1+2b)\,u_1 + (-1-2b)\,u_3 + (-1-2b)\,u_5 + (1+2b)\,u_7 = 0.$$

(2) $\underline{4,2,6,8}(1212)$:

$$(1+2b)\,u_2 + (-1-2b)\,u_4 + (-1-2b)\,u_6 + (1+2b)\,u_8 = 0.$$

(3) $\underline{3,1,2,4}(1010)$:

$$(-1+b)\,u_1 + (1-b)\,u_2 + (1-b)\,u_3 + (-1+b)\,u_4 = 0.$$

(4) $\underline{5,1,2,6}(2020)$:

$$(-2-b)\,u_1 + (2+b)\,u_2 + (2+b)\,u_5 + (-2-b)\,u_6 = 0.$$

(5) $\underline{7,5,1,2,6,8}(120210)$:

$$\begin{aligned}
&(-2-b)\,u_1 + (2+b)\,u_2 + (1+2b)\,u_5 + (-1-2b)\,u_6\\
&\quad + (1-b)\,u_7 + (-1+b)\,u_8 = 0.
\end{aligned}$$

After deleting zero equations and row-reducing the remaining system in the column order $(u_1,\dots,u_8)$, we obtain

$$A_{217} = \begin{bmatrix}
1 & 0 & 0 & -1 & 0 & -1 & -1 & 2\\
0 & 1 & 0 & -1 & 0 & -1 & 0 & 1\\
0 & 0 & 1 & -1 & 0 & 0 & -1 & 1\\
0 & 0 & 0 & 0 & 1 & -1 & -1 & 1
\end{bmatrix}.$$

Hence this is the RREF of the fundamental-cycle system.

6.2 RREF solution and conditions *(b)* and *(d)*

The free variables are

$$u_4,\ u_6,\ u_7,\ u_8.$$

The pivot variables are therefore

$$\begin{aligned} u_1 &= u_4 + u_6 + u_7 - 2u_8, \\ u_2 &= u_4 + u_6 - u_8, \\ u_3 &= u_4 + u_7 - u_8, \\ u_5 &= u_6 + u_7 - u_8. \end{aligned}$$

Imposing $u_1 = 1$ gives the main congruence

$$-1 + u_4 + u_6 + u_7 - 2u_8 = 0,$$

which is condition *(b)* after substituting $u_i = a^{t_i}$. Requiring the remaining dependent variables above to belong to $\langle a \rangle$ gives condition *(d)*.

6.3 Distinctness conditions *(a)* and *(c)*

For the distinctness calculation we solve the main congruence for u_7:

$$u_7 = 1 - u_4 - u_6 + 2u_8.$$

After this substitution, the common x-coordinate of each block of M_0 is

$$\begin{aligned} x_1 = x_2 &= 0, \\ x_3 = x_4 &= (-1+b)\,u_6 + (1-b)\,u_8, \\ x_5 = x_6 &= (-2-b)\,u_4 + (2+b)\,u_8, \\ x_7 = x_8 &= (-2-b)\,u_4 + (-1+b)\,u_6 + 3u_8. \end{aligned}$$

Hence different level-0 blocks have distinct x-coordinates, and the columns inside each block are distinct, exactly when the following factors are all nonzero (factors differing by a nonzero scalar are identified):

$$\begin{aligned} u_6 - u_8 &\not\equiv 0 \pmod m, \\ u_4 - u_8 &\not\equiv 0 \pmod m, \\ u_4 - bu_6 + (-1+b)\,u_8 &\not\equiv 0 \pmod m, \\ u_4 + bu_6 + (-1-b)\,u_8 &\not\equiv 0 \pmod m, \\ 1 - u_4 - u_6 + u_8 &\not\equiv 0 \pmod m. \end{aligned}$$

Using the main congruence, these factors reduce to condition *(a)* together with the nonzero factors in condition *(c)*. Explicitly,

$$\begin{aligned} &\text{a1:}\ \ t_4 \neq t_8 \iff u_4 - u_8 \not\equiv 0 \pmod m, \\ &\text{a2:}\ \ t_6 \neq t_8 \iff u_6 - u_8 \not\equiv 0 \pmod m, \\ &\text{a3:}\ \ t_7 \neq t_8 \iff 1 - u_4 - u_6 + u_8 \not\equiv 0 \pmod m, \\ &\qquad\ \ \text{c1} \iff u_4 + bu_6 + (-1-b)\,u_8 \not\equiv 0 \pmod m, \\ &\qquad\ \ \text{c2} \iff u_4 - bu_6 + (-1+b)\,u_8 \not\equiv 0 \pmod m. \end{aligned}$$

Hence conditions *(a)* and *(c)* are precisely the level-0 and within-block distinctness requirements for this matching configuration. Together with conditions *(b)* and *(d)*, they give the necessary and sufficient algebraic conditions for this case.

7 Case VI: Subcase 2-3-5

7.1 Matching configuration and fundamental-cycle system

The matching set is

$$M_0 = \{\{c_1, c_2\},\ \{c_3, c_4\},\ \{c_5, c_6\},\ \{c_7, c_8\}\},$$
$$M_1 = \{\{c_1, c_3\},\ \{c_2, c_6\},\ \{c_5, c_7\},\ \{c_4, c_8\}\},$$
$$M_2 = \{\{c_1, c_5\},\ \{c_2, c_4\},\ \{c_3, c_7\},\ \{c_6, c_8\}\}.$$

We choose the spanning tree

$$F = \{c_1, c_3\}_1,\ \{c_2, c_6\}_1,\ \{c_4, c_8\}_1,\ \{c_5, c_7\}_1,\ \{c_1, c_5\}_2,\ \{c_2, c_4\}_2,\ \{c_1, c_2\}_0.$$

This tree has 7 edges. Since the matching graph has 12 edges, it determines 5 fundamental cycles. The associated cycle equations are listed below.

(1) $\underline{3, 1, 5, 7}(1212)$:

$$(1 + 2b)\, u_1 + (-1 - 2b)\, u_3 + (-1 - 2b)\, u_5 + (1 + 2b)\, u_7 = 0.$$

(2) $\underline{6, 2, 4, 8}(1212)$:

$$(1 + 2b)\, u_2 + (-1 - 2b)\, u_4 + (-1 - 2b)\, u_6 + (1 + 2b)\, u_8 = 0.$$

(3) $\underline{3, 1, 2, 4}(1020)$:

$$(-1 + b)\, u_1 + (2 + b)\, u_2 + (1 - b)\, u_3 + (-2 - b)\, u_4 = 0.$$

(4) $\underline{5, 1, 2, 6}(2010)$:

$$(-2 - b)\, u_1 + (1 - b)\, u_2 + (2 + b)\, u_5 + (-1 + b)\, u_6 = 0.$$

(5) $\underline{7, 5, 1, 2, 4, 8}(120210)$:

$$\begin{aligned}(-2 - b)\, u_1 + (2 + b)\, u_2 + (-1 - 2b)\, u_4 + (1 + 2b)\, u_5 \\ + (1 - b)\, u_7 + (-1 + b)\, u_8 = 0.\end{aligned}$$

After deleting zero equations and row-reducing the remaining system in the column order $(u_1, \ldots, u_8)$, we obtain

$$A_{235} = \begin{bmatrix} 1 & 0 & 0 & b & 0 & -1-b & -1 & 1 \\ 0 & 1 & 0 & -1 & 0 & -1 & 0 & 1 \\ 0 & 0 & 1 & b & 0 & 0 & -1 & -b \\ 0 & 0 & 0 & 0 & 1 & -1-b & -1 & 1+b \end{bmatrix}.$$

Hence this is the RREF of the fundamental-cycle system.

7.2 RREF solution and conditions *(b)* and *(d)*

The free variables are

$$u_4,\ u_6,\ u_7,\ u_8.$$

The pivot variables are therefore

$$\begin{aligned}
u_1 &= -bu_4 + (1+b)\,u_6 + u_7 - u_8,\\
u_2 &= u_4 + u_6 - u_8,\\
u_3 &= -bu_4 + u_7 + bu_8,\\
u_5 &= (1+b)\,u_6 + u_7 + (-1-b)\,u_8.
\end{aligned}$$

Imposing $u_1 = 1$ gives the main congruence

$$-1 - bu_4 + (1+b)\,u_6 + u_7 - u_8 = 0,$$

which is condition *(b)* after substituting $u_i = a^{t_i}$. Requiring the remaining dependent variables above to belong to $\langle a \rangle$ gives condition *(d)*.

7.3 Distinctness conditions *(a)* and *(c)*

For the distinctness calculation we solve the main congruence for u_4:

$$u_4 = 1 + b - bu_6 + (-1-b)\,u_7 + (1+b)\,u_8.$$

After this substitution, the common x-coordinate of each block of M_0 is

$$\begin{aligned}
x_1 &= x_2 = 0,\\
x_3 &= x_4 = (-2-b)\,u_6 + (2+b)\,u_8,\\
x_5 &= x_6 = -2 - b + (1+2b)\,u_6 + (2+b)\,u_7 + (-1-2b)\,u_8,\\
x_7 &= x_8 = -2 - b + (-1+b)\,u_6 + (2+b)\,u_7 + (1-b)\,u_8.
\end{aligned}$$

Hence different level-0 blocks have distinct x-coordinates, and the columns inside each block are distinct, exactly when the following factors are all nonzero (factors differing by a nonzero scalar are identified):

$$\begin{gathered}
u_6 - u_8 \not\equiv 0 \pmod m,\\
1 + (-1-b)\,u_6 - u_7 + (1+b)\,u_8 \not\equiv 0 \pmod m,\\
1 - bu_6 - u_7 + bu_8 \not\equiv 0 \pmod m,\\
1 + (-2-b)\,u_6 - u_7 + (2+b)\,u_8 \not\equiv 0 \pmod m,\\
1 + (-2-b)\,u_6 + bu_7 + u_8 \not\equiv 0 \pmod m,\\
1 + (-1-b)\,u_6 + bu_7 \not\equiv 0 \pmod m,\\
u_6 + (-1-b)\,u_7 + bu_8 \not\equiv 0 \pmod m,\\
u_7 - u_8 \not\equiv 0 \pmod m.
\end{gathered}$$

Using the main congruence, these factors reduce to condition *(a)* together with the nonzero factors in condition *(c)*. Explicitly,

$$\begin{aligned}
&\text{a1:}\quad t_4 \neq t_8 \iff 1 + (-1-b)\,u_6 - u_7 + (1+b)\,u_8 \not\equiv 0 \pmod m,\\
&\text{a2:}\quad t_6 \neq t_8 \iff u_6 - u_8 \not\equiv 0 \pmod m,\\
&\text{a3:}\quad t_7 \neq t_8 \iff u_7 - u_8 \not\equiv 0 \pmod m,\\
&\qquad\text{c1} \iff 1 + (-2-b)\,u_6 + bu_7 + u_8 \not\equiv 0 \pmod m,\\
&\qquad\text{c2} \iff 1 + (-1-b)\,u_6 + bu_7 \not\equiv 0 \pmod m,\\
&\qquad\text{c3} \iff u_6 + (-1-b)\,u_7 + bu_8 \not\equiv 0 \pmod m,\\
&\qquad\text{c4} \iff 1 + (-2-b)\,u_6 - u_7 + (2+b)\,u_8 \not\equiv 0 \pmod m,\\
&\qquad\text{c5} \iff 1 - bu_6 - u_7 + bu_8 \not\equiv 0 \pmod m.
\end{aligned}$$

Hence conditions *(a)* and *(c)* are precisely the level-0 and within-block distinctness requirements for this matching configuration. Together with conditions *(b)* and *(d)*, they give the necessary and sufficient algebraic conditions for this case.

8 Case VII: Subcase 2-4-3

8.1 Matching configuration and fundamental-cycle system

The matching set is

$$\begin{aligned} M_0 &= \{\{c_1, c_2\},\ \{c_3, c_4\},\ \{c_5, c_6\},\ \{c_7, c_8\}\}, \\ M_1 &= \{\{c_1, c_3\},\ \{c_2, c_7\},\ \{c_4, c_5\},\ \{c_6, c_8\}\}, \\ M_2 &= \{\{c_1, c_5\},\ \{c_2, c_6\},\ \{c_3, c_7\},\ \{c_4, c_8\}\}. \end{aligned}$$

We choose the spanning tree

$$F = \{c_1, c_3\}_1,\ \{c_2, c_7\}_1,\ \{c_4, c_5\}_1,\ \{c_6, c_8\}_1,\ \{c_1, c_5\}_2,\ \{c_2, c_6\}_2,\ \{c_3, c_7\}_2.$$

This tree has 7 edges. Since the matching graph has 12 edges, it determines 5 fundamental cycles. The associated cycle equations are listed below.

(1) $\underline{4, 5, 1, 3, 7, 2, 6, 8}(12121212)$:

$$\begin{aligned} &(-1-2b)\,u_1 + (1+2b)\,u_2 + (1+2b)\,u_3 + (-1-2b)\,u_4 \\ &\quad + (1+2b)\,u_5 + (-1-2b)\,u_6 + (-1-2b)\,u_7 + (1+2b)\,u_8 = 0. \end{aligned}$$

(2) $\underline{1, 3, 7, 2}(1210)$:

$$(1-b)\,u_1 + (-1+b)\,u_2 + (1+2b)\,u_3 + (-1-2b)\,u_7 = 0.$$

(3) $\underline{3, 1, 5, 4}(1210)$:

$$(1+2b)\,u_1 + (1-b)\,u_3 + (-1+b)\,u_4 + (-1-2b)\,u_5 = 0.$$

(4) $\underline{5, 1, 3, 7, 2, 6}(212120)$:

$$\begin{aligned} &(-1-2b)\,u_1 + (1+2b)\,u_2 + (1+2b)\,u_3 + (2+b)\,u_5 \\ &\quad + (-2-b)\,u_6 + (-1-2b)\,u_7 = 0. \end{aligned}$$

(5) $\underline{7, 2, 6, 8}(1210)$:

$$(1+2b)\,u_2 + (-1-2b)\,u_6 + (1-b)\,u_7 + (-1+b)\,u_8 = 0.$$

After deleting zero equations and row-reducing the remaining system in the column order $(u_1, \ldots, u_8)$, we obtain

$$A_{243} = \begin{bmatrix} 1 & 0 & 0 & 0 & -1 & 0 & -1-b & 1+b \\ 0 & 1 & 0 & 0 & 0 & -1 & -1-b & 1+b \\ 0 & 0 & 1 & 0 & -1-b & 1+b & -1 & 0 \\ 0 & 0 & 0 & 1 & -1-b & 1+b & 0 & -1 \end{bmatrix}.$$

Hence this is the RREF of the fundamental-cycle system.

8.2 RREF solution and conditions *(b)* and *(d)*

The free variables are

$$u_5,\ u_6,\ u_7,\ u_8.$$

The pivot variables are therefore

$$\begin{aligned}
u_1 &= u_5 + (1+b)\,u_7 + (-1-b)\,u_8,\\
u_2 &= u_6 + (1+b)\,u_7 + (-1-b)\,u_8,\\
u_3 &= (1+b)\,u_5 + (-1-b)\,u_6 + u_7,\\
u_4 &= (1+b)\,u_5 + (-1-b)\,u_6 + u_8.
\end{aligned}$$

Imposing $u_1 = 1$ gives the main congruence

$$-1 + u_5 + (1+b)\,u_7 + (-1-b)\,u_8 = 0,$$

which is condition *(b)* after substituting $u_i = a^{t_i}$. Requiring the remaining dependent variables above to belong to $\langle a \rangle$ gives condition *(d)*.

8.3 Distinctness conditions *(a)* and *(c)*

For the distinctness calculation we solve the main congruence for u_5:

$$u_5 = 1 + (-1-b)\,u_7 + (1+b)\,u_8.$$

After this substitution, the common x-coordinate of each block of M_0 is

$$\begin{aligned}
x_1 &= x_2 = 0,\\
x_3 &= x_4 = 1 + 2b + (-2-b)\,u_6 - 3bu_7 + (1+2b)\,u_8,\\
x_5 &= x_6 = (-1-2b)\,u_7 + (1+2b)\,u_8,\\
x_7 &= x_8 = (-1+b)\,u_6 + (-1-2b)\,u_7 + (2+b)\,u_8.
\end{aligned}$$

Hence different level-0 blocks have distinct x-coordinates, and the columns inside each block are distinct, exactly when the following factors are all nonzero (factors differing by a nonzero scalar are identified):

$$\begin{gathered}
1 + bu_6 + (-2-b)\,u_7 + u_8 \not\equiv 0 \pmod m,\\
u_7 - u_8 \not\equiv 0 \pmod m,\\
u_6 + bu_7 + (-1-b)\,u_8 \not\equiv 0 \pmod m,\\
1 + bu_6 + (-1-b)\,u_7 \not\equiv 0 \pmod m,\\
1 - u_6 + (-1-b)\,u_7 + (1+b)\,u_8 \not\equiv 0 \pmod m,\\
u_6 - u_8 \not\equiv 0 \pmod m.
\end{gathered}$$

Using the main congruence, these factors reduce to condition *(a)* together with the nonzero factors in condition *(c)*. Explicitly,

$$\begin{aligned}
&\text{a1:}\ \ t_5 \neq t_6 \iff 1 - u_6 + (-1-b)\,u_7 + (1+b)\,u_8 \not\equiv 0 \pmod m,\\
&\text{a2:}\ \ t_6 \neq t_8 \iff u_6 - u_8 \not\equiv 0 \pmod m,\\
&\text{a3:}\ \ t_7 \neq t_8 \iff u_7 - u_8 \not\equiv 0 \pmod m,\\
&\qquad\ \text{c1} \iff 1 + bu_6 + (-2-b)\,u_7 + u_8 \not\equiv 0 \pmod m,\\
&\qquad\ \text{c2} \iff u_6 + bu_7 + (-1-b)\,u_8 \not\equiv 0 \pmod m,\\
&\qquad\ \text{c3} \iff 1 + bu_6 + (-1-b)\,u_7 \not\equiv 0 \pmod m,\\
&\qquad\ \text{c4} \iff 1 - u_6 + (-1-b)\,u_7 + (1+b)\,u_8 \not\equiv 0 \pmod m.
\end{aligned}$$

Hence conditions *(a)* and *(c)* are precisely the level-0 and within-block distinctness requirements for this matching configuration. Together with conditions *(b)* and *(d)*, they give the necessary and sufficient algebraic conditions for this case.

9 Conclusion

For Cases II–VII, the fundamental-cycle systems yield the displayed RREFs; with $u_1 = 1$, their pivot relations give conditions *(b)* and *(d)*, while the exact level-0 partition and pairwise-distinctness tests give conditions *(a)* and *(c)*. Thus the six alternatives in Theorem 36 are fully verified.